

Federated Variational Preference Alignment with Gumbel-Softmax Prior for Personalized User Preferences

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Abstract

Existing Federated Learning (FL) frameworks for LLM alignment rely on monolithic reward models, failing to capture inherently conflicting user preferences (e.g., *helpfulness* vs. *harmlessness*). While Variational Preference Learning (VPL) addresses heterogeneity, its application to FL is hindered by local data sparsity, which destabilizes posterior estimation. We propose Federated Variational Preference Alignment with Gumbel-Softmax Prior (FedVPA-GP), a framework designed to capture diverse preferences in decentralized settings. To stabilize inference, we introduce a Federated Mixture Prior that allows clients to leverage global knowledge without sharing raw data. Furthermore, we employ difference embeddings with an orthogonal loss to explicitly enforce the separation of preference prototypes in the latent space. Experiments on HH-RLHF and Fed-ChatbotPA demonstrate that FedVPA-GP significantly outperforms monolithic baselines and effectively switches between response modes based on inferred user contexts.

1. Introduction

2. Introduction

Reinforcement Learning from Human Feedback (RLHF) has established itself as the standard for aligning LLMs (Ziegler et al., 2019). However, its reliance on centralized data poses a fundamental bottleneck: high-quality preference data—reflecting personal, cultural, and political nuances—resides on edge devices, where centralization risks privacy violations and GDPR non-compliance (European Parliament and Council of the European Union, 2016).

Federated Learning (FL) offers a privacy-preserving alterna-

tive. Frameworks like *FedBiscuit* (Wu et al., 2024) successfully mitigate computational costs by training lightweight modules on frozen LLMs. However, these frameworks suffer from a critical limitation: they enforce a monolithic reward model. Recent research highlights that user preferences are inherently conflicting (Santurkar et al., 2023), most notably between *helpfulness* and *harmlessness* (Poddar et al., 2024), or on stylistic traits like verbosity (Lee et al., 2024). Aggregating these heterogeneous distributions into a single proxy yields theoretical sub-optimality (Shirali et al., 2025), resulting in a “one-size-fits-all” model that fails to satisfy distinct clients.

While *Variational Preference Learning* (VPL) (Poddar et al., 2024) addresses heterogeneity by conditioning on latent variables (z), adapting it to FL is non-trivial. Unlike centralized settings where models learn the global manifold from pooled data, FL clients operate in isolation. Due to *preference heterogeneity* and *data sparsity*, clients lack the global context to position their preferences, leading to unstable posterior estimation when learning from limited local samples.

To overcome these challenges, we propose Federated Variational Preference Alignment with Gumbel-Softmax Prior (FedVPA-GP). Instead of a fixed Gaussian prior, we introduce a Federated Mixture Prior that aggregates learned distributions across the federation. This approach allows clients to effectively account for the distributions of other clients, leveraging global knowledge to stabilize variational inference even amidst conflicting and diverse preferences. To enable end-to-end training, we employ the Gumbel-Softmax relaxation (Jang et al., 2017). Furthermore, to ensure the latent variable z accurately captures preference semantics, we incorporate difference embeddings guided by an orthogonal loss. This mechanism enforces the separation of distinct preference prototypes, preventing mode collapse.

Extensive experiments on the HH-RLHF and FedChatbotPA datasets demonstrate that FedVPA-GP significantly outperforms monolithic baselines (e.g., FedBiscuit, FedDPO). Qualitative analysis further confirms that our algorithm successfully disentangles preferences in the latent space, enabling the model to dynamically switch between “helpful” and “harmless” modes based on the inferred z .

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Federated Variational Preference Alignment (FedVPA)

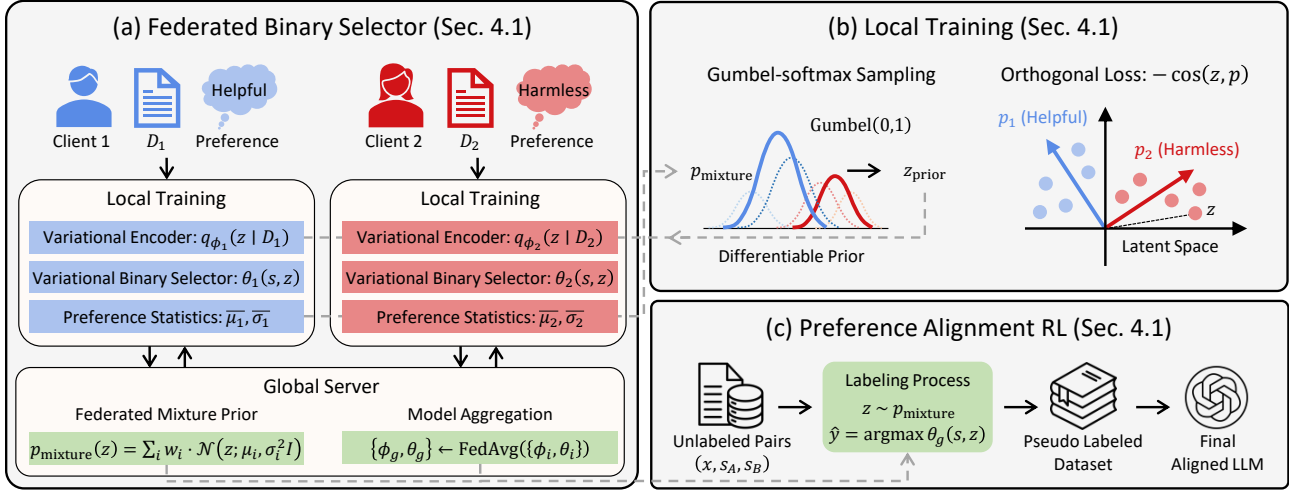


Figure 1. Temp

Our contributions are summarized as follows:

- **Federated Variational Preference Alignment (FedVPA-GP):** We propose the first federated framework addressing conflicting user preferences. By adapting variational inference to FL, we resolve the theoretical sub-optimality of monolithic reward models.
- **Stabilized Inference and Disentanglement:** To mitigate data sparsity and separate conflicting preferences, we introduce a Federated Mixture Prior with Gumbel-Softmax and an orthogonal loss with difference embeddings. This combined approach stabilizes posterior estimation and variational inference.
- **Empirical Validation:** Experiments on HH-RLHF and FedChatbotPA demonstrate that FedVPA-GP significantly outperforms monolithic baselines and enables dynamic preference switching based on the inferred context.

3. Related Works

Reinforcement Learning from Human Feedback (RLHF) Standard RLHF aligns LLMs via reward modeling (Ziegler et al., 2019; Stiennon et al., 2020) or, more recently, direct preference optimization (DPO) without explicit rewards (Rafailov et al., 2023). However, these centralized approaches necessitate massive data aggregation, creating significant privacy risks and scalability bottlenecks (Bai et al., 2022).

Federated Preference Alignment To mitigate privacy risks, Federated Learning has been adapted for alignment.

(Wu et al., 2024) proposed *FedBis* and *FedBiscuit* to approximate global rewards using client-side binary selectors, while Ye et al. (2024) introduced *FedDPO* for distributed direct policy optimization. Despite enabling privacy-preserving training, these frameworks prioritize a *global consensus* or coarse-grained clusters. By averaging out conflicting views (e.g., helpfulness vs. harmlessness), they fail to capture the continuous heterogeneity of individual preferences, resulting in a suboptimal “one-size-fits-all” model (Poddar et al., 2024).

Personalized Preference Learning Recent studies challenge monolithic alignment, showing that preferences often conflict (Santurkar et al., 2023) across diverse axes like verbosity (Lee et al., 2024), where simple aggregation leads to theoretical sub-optimality (Shirali et al., 2025). To enable personalization, prior works employ weight merging (Jang et al., 2023), multi-objective interpolation (Rame et al., 2023), or latent variables via *Variational Preference Learning* (VPL) (Poddar et al., 2024). However, these centralized methods rely on global data access, making them incompatible with the data silos and privacy constraints of Federated Learning. We address this gap by adapting variational personalization to the decentralized setting.

4. Preliminaries

We consider a Federated Learning (FL) system consisting of K clients. Each client $i \in \{1, \dots, K\}$ possesses a private dataset of pairwise preferences $\mathcal{D}_i = \{(s_A, s_B, y)\}_{j=1}^{N_i}$, where s_A and s_B are two candidate responses from a Large Language Model (LLM), and $y \in \{0, 1\}$ indicates the user’s preference (with $y = 1$ denoting $s_A \succ s_B$).

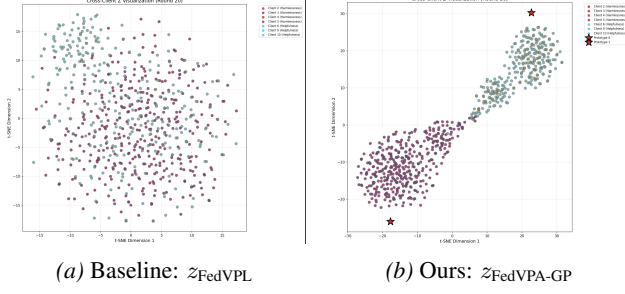


Figure 2. **Visualization of Latent Variable Distributions.** (a) The baseline (FedVPL) suffers from posterior collapse, where latent codes z_{VPL} cluster indistinguishably. (b) Our method (**FedVPA-GP**) effectively disentangles user preferences, showing distinct modes in $z_{\text{FedVPA-GP}}$ corresponding to different client groups.

4.1. Standard Federated Preference Alignment

In standard Federated RLHF settings (e.g., FedDPO or FedBiscuit), the goal is to learn a global reward model $r_\theta(s_A, s_B)$ that maximizes the likelihood of user preferences across all clients (Wu et al., 2024). The preference probability is typically modeled using the Bradley-Terry-Luce (BTL) model:

$$p_\theta(y = 1 \mid s_A, s_B) = \sigma(r_\theta(s_A) - r_\theta(s_B)), \quad (1)$$

where $\sigma(\cdot)$ is the sigmoid function. The federated objective minimizes the aggregate negative log-likelihood: $\min_\theta \sum_{i=1}^K \mathbb{E}_{\mathcal{D}_i} [-\log p_\theta(y \mid s_A, s_B)]$.

However, this formulation assumes a single consensus reward function r_θ , which inevitably averages out conflicting preferences (e.g., "Helpful" vs. "Harmless") and fails to capture user-specific nuances (Poddar et al., 2024).

4.2. Variational Preference Learning

To address heterogeneity, we adopt a latent conditional framework. We assume each user i is governed by a continuous latent preference vector $z_i \in \mathbb{R}^d$ (typically $d = 32$) that conditions the reward model. For binary choice tasks, we condition the model’s logits on z_i through a learned projection network:

$$\text{logits}(s_A, s_B \mid z_i) = \text{logits}_{\text{base}}(s_A, s_B) + f_\theta(z_i), \quad (2)$$

where $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^{|C|}$ is a linear projection that maps the latent vector to logit adjustments, and C is the set of choice tokens (typically $\{A, B\}$). The choice probability is then computed via softmax over the conditioned logits.

Since z_i is unobserved, we treat it as a latent variable and employ Variational Inference (VI). We introduce a local variational posterior $q_\phi(z_i \mid \mathcal{D}_i) = \mathcal{N}(z_i; \mu_i, \sigma_i^2 I)$ parameterized by ϕ to approximate the true posterior. The encoder extracts preference features (e.g., embedding difference $\Delta h = h_{\text{chosen}} - h_{\text{rejected}}$) and outputs posterior parameters

(μ_i, σ_i^2) . The latent vector is sampled using the reparameterization trick: $z_i = \mu_i + \sigma_i \odot \epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$ and $\sigma_i = \exp(0.5 \cdot \log \sigma_i^2)$.

The objective is to maximize the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(z_i \mid \mathcal{D}_i)} [\log p_\theta(\mathcal{D}_i \mid z_i)] - \beta \mathbb{D}_{KL}(q_\phi(z_i \mid \mathcal{D}_i) \parallel p(z)), \quad (3)$$

where $p(z)$ is the prior distribution over latent preferences and β is a regularization coefficient.

4.3. Limitations of Federated Variational Preference Learning

While Variational Preference Learning (VPL) effectively captures user diversity in centralized settings, directly transposing this framework to Federated Learning introduces critical challenges stemming from *preference heterogeneity* and *data sparsity*.

Difficulty of Variational Inference under Heterogeneity: In centralized VPL, the model has access to the entire pool of user data, allowing it to implicitly learn the global structure of the latent space and the relative relationships between diverse user preferences. In contrast, FL clients perform variational inference in isolation without seeing other clients’ distributions. Since the standard Gaussian prior $\mathcal{N}(0, I)$ is agnostic to the population’s preference distribution, clients cannot position their latent embeddings z_i relative to the global manifold. This isolation makes it difficult to learn a coherent latent space where meaningful preference clusters can form.

High Variance due to Data Sparsity: In FL, the global dataset is fragmented across numerous devices, leaving individual clients with small sample sizes (N_i). This data sparsity makes the estimation of the variational posterior $q_\phi(z_i \mid \mathcal{D}_i)$ highly unstable. With insufficient samples to anchor the density estimation, the variational parameters exhibit high variance, leading to unstable gradient updates and hindering convergence.

These limitations indicate that a client cannot effectively learn a personalized preference distribution relying solely on its local data and a non-informative prior. To address this, we propose the Federated Mixture Prior (Section 5.2). By leveraging the learned preference distributions of other clients as a dynamic prior, we enable clients to consider the global preference landscape, thereby stabilizing local inference and enabling knowledge transfer despite data sparsity.

5. Main method

We propose **Federated Variational Preference Alignment with Gumbel-Softmax Prior FedVPA-GP**, a framework designed to learn personalized reward models in a privacy-

preserving manner. Unlike previous approaches that aggregate gradients into a single global model (Wu et al., 2024), FedVPA treats user personalization as a continuous latent variable inference problem. By modeling user preferences as continuous latent vectors and leveraging a federated mixture prior with Gumbel-Softmax relaxation, we achieve stable and efficient federated training while enabling knowledge sharing across clients.

5.1. Latent Conditional Preference Model

We assume that user preferences can be encoded as continuous latent vectors in a low-dimensional space. For each client i , we introduce a continuous latent variable $z_i \in \mathbb{R}^d$, where d is the latent dimension (typically $d = 32$). This latent vector captures the client’s personalized preference characteristics.

The probability that a user i prefers response s_A over s_B is conditioned on this latent code z_i via a Latent Conditional Reward Model. For binary choice tasks, we condition the model’s logits on z_i through a learned projection network:

$$\text{logits}(s_A, s_B \mid z_i) = \text{logits}_{\text{base}}(s_A, s_B) + f_\theta(z_i), \quad (4)$$

where $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{C}|}$ is a learned linear projection (latent projection) that maps the latent vector to logit adjustments, and $\mathcal{C}$ is the set of choice tokens (typically $\{A, B\}$). The choice probability is then computed via softmax over the conditioned logits.

5.2. Variational Inference with Federated Mixture Prior

Inference Network: Each client maintains a local variational encoder $q_\phi(z \mid \mathcal{D}_i)$ parameterized by ϕ . Given a preference comparison (s_A, s_B, y) , we extract hidden representations h_A, h_B from the base LLM, compute the embedding difference $\Delta h = h_{\text{chosen}} - h_{\text{rejected}}$, process it through a feature extractor $f = \text{FeatureExtractor}(\Delta h)$ (or use choice logits), and encode to posterior parameters $(\mu_i, \sigma_i^2) = \text{Encoder}(f)$. The posterior distribution is:

$$q_\phi(z \mid \mathcal{D}_i) = \mathcal{N}(z; \mu_i, \sigma_i^2 I), \quad (5)$$

where $\mu_i \in \mathbb{R}^d$ and $\log \sigma_i^2 \in \mathbb{R}^d$ are learned via a neural network. The latent vector z_i is sampled using the reparameterization trick:

$$z_i = \mu_i + \sigma_i \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (6)$$

where $\sigma_i = \exp(0.5 \cdot \log \sigma_i^2)$ and $\odot$ denotes element-wise multiplication. This ensures that gradients can flow through the sampling operation.

Preference Feature Difference: To emphasize preference signals, we compute the embedding difference $\Delta h = h_{\text{chosen}} - h_{\text{rejected}}$ (by splitting prompts at response markers), which encourages z_i to capture preference-specific

directions. The feature extractor processes Δh through a multi-layer perceptron before encoding.

Baseline VPL: In the ablation baseline, we replace the federated mixture prior with a standard Gaussian prior $p(z) = \mathcal{N}(0, I)$, leading to the KL divergence term:

$$\mathbb{D}_{KL}(q_\phi(z \mid \mathcal{D}_i) \parallel \mathcal{N}(0, I)) = \frac{1}{2} \sum_{j=1}^d (\mu_{i,j}^2 + \sigma_{i,j}^2 - \log \sigma_{i,j}^2 - 1). \quad (7)$$

Federated Mixture Prior with Gumbel-Softmax: Our default setting uses other clients’ learned preference distributions as a mixture prior. Specifically, the server collects the posterior parameters (μ_j, σ_j^2) from all participating clients in the previous round and constructs a mixture prior:

$$p_{\text{mixture}}(z) = \sum_{j=1}^{|S|} w_j \cdot \mathcal{N}(z; \mu_j, \sigma_j^2 I), \quad (8)$$

where S is the set of participating clients from the previous round and w_j are normalized weights (typically sample-size weighted: $w_j = n_j / \sum_{k \in S} n_k$).

Computing the KL divergence with this mixture prior requires evaluating:

$$\mathbb{D}_{KL}(q_\phi(z \mid \mathcal{D}_i) \parallel p_{\text{mixture}}(z)) = \mathbb{E}_{z \sim q_\phi} [\log q_\phi(z \mid \mathcal{D}_i) - \log p_{\text{mixture}}(z)], \quad (9)$$

where $\log p_{\text{mixture}}(z) = \log \sum_j w_j \cdot \mathcal{N}(z; \mu_j, \sigma_j^2 I)$ is computed using the log-sum-exp trick for numerical stability:

$$\log p_{\text{mixture}}(z) = \max_j a_j + \log \left(\sum_{j=1}^{|S|} \exp(a_j - \max_k a_k) \right), \quad (10)$$

where $a_j = \log w_j + \log \mathcal{N}(z; \mu_j, \sigma_j^2 I)$.

Gumbel-Softmax for Differentiable Prior Sampling:

For sampling (visualization/generation), we use Gumbel-Softmax relaxation (Jang et al., 2017) with temperature $\tau = 1.0$: $\alpha_j = \exp((\log w_j + g_j)/\tau) / \sum_k \exp((\log w_k + g_k)/\tau)$ where $g_j \sim \text{Gumbel}(0, 1)$, then $z_{\text{prior}} = \sum_j \alpha_j \cdot z_j$ with $z_j \sim \mathcal{N}(\mu_j, \sigma_j^2 I)$.

5.3. Federated Variational Objective (ELBO)

The training objective is to maximize the Evidence Lower Bound (ELBO) of the log-likelihood of the preference data.

For a client i , the local loss function is defined as:

$$\begin{aligned} \mathcal{L}_i(\theta, \phi) &= -\text{ELBO}(\mathcal{D}_i) \\ &= -\mathbb{E}_{z \sim q_\phi} \left[\underbrace{\sum_{(s_A, s_B, y) \in \mathcal{D}_i} \log p_\theta(y \mid s_A, s_B, z)}_{\text{Reconstruction Loss (Preference Alignment)}} \right] \\ &\quad + \beta \cdot \underbrace{\mathbb{D}_{KL}(q_\phi(z \mid \mathcal{D}_i) \parallel p_{\text{mixture}}(z))}_{\text{Regularization (Prior Matching)}} \\ &\quad + \lambda \cdot \underbrace{\mathcal{L}_{\text{orthogonal}}(z)}_{\text{Orthogonal Loss (Preference Separation)}}, \end{aligned} \quad (11)$$

The **Reconstruction Loss** encourages correct choice prediction given z (cross-entropy between conditioned logits and labels). The **Regularization term** is the KL divergence between q_ϕ and $p_{\text{mixture}}(z)$ (VPL ablation uses $p(z) = \mathcal{N}(0, I)$), preventing overfitting and enabling knowledge transfer. The **Orthogonal Loss** (optional, $\lambda \geq 0$) aligns z with preference-specific prototypes. To reduce memory, we freeze the base LLM and only update VPL components (feature extractor, variational encoder, latent projection).

5.4. Orthogonal Loss for Preference Separation

To better separate different preference patterns and suppress general features, we introduce an orthogonal loss inspired by CLOP (Li et al., 2024). For binary choice tasks, we maintain two fixed orthonormal prototype vectors $\mathbf{p}_0, \mathbf{p}_1 \in \mathbb{R}^d$ in the latent space, initialized as scaled identity vectors:

$$\mathbf{p}_0 = [s, 0, \dots, 0], \quad \mathbf{p}_1 = [0, s, \dots, 0], \quad (12)$$

where $s = 5.0$ is a scale parameter. Unlike CLOP, our prototypes are fixed (not learnable) to maintain interpretability and prevent collapse.

For each latent representation z , we automatically assign it to the closest prototype based on similarity:

$$y^* = \arg \max_{j \in \{0,1\}} z^T \mathbf{p}_j, \quad (13)$$

and encourage z to align with its assigned prototype $\mathbf{p}_{y^*}$ using the pull loss:

$$\mathcal{L}_{\text{pull}}(z) = \|z - \mathbf{p}_{y^*}\|_2^2. \quad (14)$$

To maintain orthogonality between prototypes, we add an orthonormality constraint:

$$\mathcal{L}_{\text{orthonorm}} = \|\mathbf{P}^T \mathbf{P} - \mathbf{I}\|_F^2, \quad (15)$$

where $\mathbf{P} = [\mathbf{p}_0, \mathbf{p}_1]^T$ is the prototype matrix. The combined orthogonal loss is:

$$\mathcal{L}_{\text{orthogonal}}(z) = \mathcal{L}_{\text{pull}}(z) + \gamma \cdot \mathcal{L}_{\text{orthonorm}}, \quad (16)$$

Algorithm 1 Federated Variational Preference Alignment with Mixture Prior (FedVPA)

Require: Clients K , rounds T , local steps E , learning rate η , KL weight β , orthogonal weight λ
Ensure: Global parameters θ^T, ϕ^T , client average z distributions $\{\bar{\mu}_i^T, \bar{\sigma}_i^2\}_{i=1}^K$

- 1: Initialize θ^0, ϕ^0 (base model frozen, only VPL components trainable)
- 2: **for** round $t = 1, 2, \dots, T$ **do**
- 3: **Server:** Sample clients $\mathcal{S}^t \subseteq \{1, \dots, K\}$ (typically $|\mathcal{S}^t| = 10$)
- 4: **Server:** Broadcast θ^t, ϕ^t to $\mathcal{S}^t$
- 5: **if** $t > 1$ **then**
- 6: **Server:** Broadcast mixture prior $\{(\mu_j, \sigma_j^2), w_j\}_{j \in \mathcal{S}^{t-1}}$ to $\mathcal{S}^t$
- 7: **end if**
- 8: **Parallel on clients** $i \in \mathcal{S}^t$:
- 9: **for** client $i \in \mathcal{S}^t$ **in parallel do**
- 10: Receive θ^t, ϕ^t ; Update local prior $p_{\text{mixture}}(z)$ if received
- 11: Initialize $\theta_i^t \leftarrow \theta^t, \phi_i^t \leftarrow \phi^t$
- 12: **for** local step $e = 1, \dots, E$ **do**
- 13: **for** batch $(s_A, s_B, y) \in \mathcal{D}_i$ **do**
- 14: Extract features: $h_{\text{chosen}}, h_{\text{rejected}} \leftarrow \text{LLM}(s_A, s_B)$
- 15: Compute difference: $\Delta h = h_{\text{chosen}} - h_{\text{rejected}}$ (or use choice logits)
- 16: Process: $f_i \leftarrow \text{FeatureExtractor}(\Delta h)$
- 17: Encode: $(\mu_i, \sigma_i^2) \leftarrow q_{\phi_i^t}(f_i)$
- 18: Sample: $z_i \sim \mathcal{N}(\mu_i, \sigma_i^2 I)$ via reparameterization
- 19: Project: $\Delta \text{logits} \leftarrow \text{LatentProjection}(z_i)$
- 20: Condition: $\text{logits} \leftarrow \text{logits}_{\text{base}} + \Delta \text{logits}$
- 21: Compute: $\mathcal{L}_i \leftarrow -\log p_{\theta_i^t}(y \mid s_A, s_B, z_i) + \beta \cdot \mathbb{D}_{KL}(q_{\phi_i^t}(z \mid \cdot) \parallel p_{\text{mixture}}(z)) + \lambda \cdot \mathcal{L}_{\text{orthogonal}}(z_i)$
- 22: Update: θ_i^t, ϕ_i^t via SGD on $\mathcal{L}_i$ (only VPL components, base model frozen)
- 23: **end for**
- 24: **end for**
- 25: Compute average: $\bar{\mu}_i, \bar{\sigma}_i^2$ over local batches
- 26: Send $(\theta_i^t, \phi_i^t, \bar{\mu}_i, \bar{\sigma}_i^2, |\mathcal{D}_i|)$ to server
- 27: **end for**
- 28: **Server:** Aggregate: $\theta^{t+1} \leftarrow \frac{1}{|\mathcal{S}^t|} \sum_{i \in \mathcal{S}^t} \theta_i^t, \phi^{t+1} \leftarrow \frac{1}{|\mathcal{S}^t|} \sum_{i \in \mathcal{S}^t} \phi_i^t$
- 29: **Server:** Collect z-distributions $\{(\bar{\mu}_i, \bar{\sigma}_i^2, n_i)\}_{i \in \mathcal{S}^t}$
- 30: **Server:** Compute weights $w_i \leftarrow n_i / \sum_{j \in \mathcal{S}^t} n_j$ and store mixture prior
- 31: **end for**

where $\gamma = 0.1$ is a weight parameter. This loss encourages latent representations z to cluster around their nearest prototypes while ensuring that different preference types occupy orthogonal subspaces, thereby suppressing general features that are not preference-specific.

Table 1. **Main Results on HH-RLHF.** We report the GPT-4 Win-rate (%) across varying client counts ($N \in \{10, 50, 100\}$). Shaded rows indicate our proposed method, **FedVPA-GP**, which consistently achieves the best trade-off between helpfulness and harmlessness.

MODEL	METHOD	10 CLIENTS		50 CLIENTS		100 CLIENTS	
		WIN-RATE (%)		WIN-RATE (%)		WIN-RATE (%)	
		HELPFUL	HARMLESS	HELPFUL	HARMLESS	HELPFUL	HARMLESS
QWEN 2	FEDDPO	42.5	58.1	40.2	55.4	38.5	53.2
	FEDBISCUIT	46.8	62.3	44.5	60.1	41.3	57.8
	FEDVPL	49.2	65.5	47.8	63.2	45.6	60.4
	FEDVPA-GP	54.1	71.2	53.5	70.4	52.8	69.1
GEMMA-2B	FEDDPO						
	FEDBISCUIT						
	FEDVPL						
	FEDVPA-GP						

6. Experiments

6.1. Experimental Settings

6.1.1. DATASETS

We evaluate our method on the Fed-ChatbotPA dataset, a federated preference alignment benchmark designed for personalized chatbot training. The dataset consists of pairwise preference comparisons across multiple clients, where each client has distinct preference patterns. The dataset is split into train (90%), validation (9%), and test (1%) sets using a meta-splitter to ensure realistic federated learning scenarios.

6.1.2. MODELS

We conduct experiments using two base language models to demonstrate the scalability of our approach:

- **Qwen-2 0.5B:** A compact 0.5 billion parameter model from the Qwen-2(Yang et al., 2024) family, suitable for resource-constrained federated environments.
- **Gemma-2B:** A 2 billion parameter model from Google’s Gemma family(Gemma Team et al., 2024), providing a larger model baseline for comparison.

Both models are fine-tuned using LoRA (Low-Rank Adaptation) (?) to enable efficient parameter-efficient fine-tuning in federated settings. We use LoRA rank $r = 8$, LoRA alpha $\alpha = 16$, and dropout rate $p = 0.05$.

6.1.3. BASELINES

We compare our proposed **FedVPA** framework with the following baselines:

- **FedBiscuit:** Federated learning with multiple LoRA adapters (U=3)? for personalization (Wu et al., 2024).

- **FedDPO:** Federated Direct Preference Optimization (Ye et al., 2024).
- **VPL:** Our baseline Variational Preference Learning with standard Gaussian prior $\mathcal{N}(0, I)$.
- **VPL-GP:** Our proposed method with federated mixture prior and Gumbel-Softmax relaxation.

6.2. Personalization in Federated Preference Alignment

Main Experiment Todo...

6.3. Analysis

z distribution analysis Todo...

example Todo...

adaptation

6.4. Ablation

algorithm ablation Todo...

real client prompt experiment

privacy Todo...

7. Conclusion

In this work, we identified the limitations of monolithic reward models in Federated Learning, which fail to address the inherent heterogeneity of user preferences. To bridge this gap, we introduced **FedVPA-GP**, a novel framework that adapts variational preference learning to decentralized settings. By incorporating a Federated Mixture Prior with

Gumbel-Softmax relaxation, our method stabilizes variational inference against local data sparsity, enabling clients to leverage collective knowledge without compromising privacy. Furthermore, our use of difference embeddings and orthogonal loss ensures the effective disentanglement of conflicting preference prototypes, such as *helpfulness* and *harmlessness*.

Empirical results on HH-RLHF and FedChatbotPA demonstrate that FedVPA-GP not only outperforms existing monolithic baselines but also successfully generates responses tailored to distinct user needs through latent variable control. These findings suggest that personalized alignment in federated systems is not only feasible but essential for deploying LLMs that respect diverse human values. Future work will explore applying FedVPA-GP to larger-scale heterogeneous client distributions and investigating its robustness against adversarial attacks in open federated networks.

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A. Appenix A

A.0.1. HYPERPARAMETERS

For all experiments, we use the following hyperparameter settings unless otherwise specified:

Federated Learning Settings:

- Number of clients: $K = 10$
- Clients sampled per round: $|\mathcal{S}| = 5$
- Total communication rounds: $T = 50$
- Local update steps per round: $E = 30$
- Batch size: $B = 4$
- Gradient accumulation steps: 2 (effective batch size = 8)

Optimization:

- Optimizer: AdamW (?)
- Learning rate: $\eta = 1 \times 10^{-5}$
- Beta parameters: $\beta_1 = 0.9, \beta_2 = 0.95$
- Half-precision training: Enabled (FP16)

VPL-GP Specific Hyperparameters:

- Latent dimension: $d = 32$
- KL divergence weight: $\beta = 0.1$ (unless specified otherwise in hyperparameter search)
- Gumbel-Softmax temperature: $\tau = 1.0$
- Feature extraction method: Choice logits (concatenation of chosen and rejected response logits)

Hyperparameter Search: We conduct a comprehensive hyperparameter search over the following ranges:

- VPL KL weight: $\beta \in \{0.001, 0.01, 0.05, 0.1\}$
- Learning rate: $\eta \in \{1 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$
- Local update steps: $E \in \{30, 40, 50, 60\}$

This results in 64 total combinations, from which we select the best configuration based on validation performance.

A.0.2. EVALUATION METRICS

We evaluate model performance using the following metrics:

- **Accuracy:** Binary classification accuracy on preference prediction tasks.
- **Loss:** Cross-entropy loss on the test set.
- **KL Divergence:** Measures the divergence between posterior and prior distributions, indicating how well the model utilizes the latent space.
- **Reconstruction Loss:** The negative log-likelihood term in ELBO, measuring preference prediction quality.
- **Convergence Speed:** Number of rounds required to reach a target accuracy threshold.

A.0.3. IMPLEMENTATION DETAILS

All experiments are implemented using PyTorch (?) and the FederatedScope framework (?). We use LoRA adapters from the PEFT library (?) for parameter-efficient fine-tuning. Training is performed on NVIDIA GPUs with mixed-precision (FP16) to reduce memory consumption. We report results averaged over 3 random seeds for statistical significance.